University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Digital Logic Design	**Course Code:** CMPE-222L
Assignment Type: Lab	**Dated:** 29th September 2026
Semester: 3rd	**Session:** 2025
Lab Number: 2	**CLOs to be covered: CLO 1, CLO 2, CLO 3**
Lab Title: Signed and unsigned numbers	**Teacher Name:** Fareed Jafri

Lab Goals

- Signed and unsigned addition of 4-bit binary numbers using an integrated circuit (CLO1, CLO3).
- Signed subtraction of 4-bit binary numbers (2's complement) using an integrated circuit (CLO1, CLO3).
- Interface a seven segment display unit with a 4-bit binary subtractor (CLO1).

Equipment Required

- **ICs:** 7483 (1x), 7485 (1x)
- **Resistors:** 220 Ohms (7x)
- **Tools:** IC puller (1x), Nose pliers (1x), Wire stripper (1x)
- **Other:** Jumper wires (as needed)

Lab Evaluation

	CLOs	Maximum marks	Marks obtained
Task 1a	1	2	
Task 1b	3	2	
Task 2a	2	1	
Task 2b	1	2	
Task 2c	3	1	
Task 3	1	2	
Total			**/10**

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric	CLOs
Completed	**100%**	L1	Completed the task correctly.	All
Great attempt	**75%**	L2	Almost completed the task, with minor deficiencies	2,3
Good attempt	**50%**	L3	Did not complete the task but managed to reasonably attempt it.	All
Basic attempt	**25%**	L4	Did not complete the task but managed to basically attempt it.	All

Did not attempt	**0%**	L5	Did not complete the task or attempt to do it.	All

Signed and unsigned addition of 4-bit binary numbers using an integrated circuit

Task 1a [CLO1, 2 Marks]

In this task you will be doing **signed** and **unsigned** addition of binary numbers. The 7483 IC is a 4-bit binary adder, which can add two signed 4-bit binary numbers A and B to produce a signed 4-bit output C. The pinout of the IC is shown in the 'Pin out' section at the end. Wire up the IC as follows. Switches S0 to S3 provide the first signed 4-bit binary input A = $A_3A_2A_1A_0$. Switches S4 to S7 provide the second 4-bit binary output B = $B_3B_2B_1B_0$. LEDs L0 to L3 record the 4-bit binary output C = $C_3C_2C_1C_0$. What needs to be done to C_{in}?

Task 1b [CLO3, 2 Marks]

Now fill in the following table using the circuit from Task 1a. Each 4-bit number is an **unsigned** number. Circle those binary outputs where **unsigned** overflow has occurred.

Input A		**Input B**		**Output C = A + B**	
Binary	**Decimal**	**Binary**	**Decimal**	**Binary**	**Decimal**
0011		0000			
0011		0001			
0011		0010			
0011		0011			
0011		0100			
0011		0101			
0011		0110			
0011		0111			
1000		0101			
1001		0101			
1010		0101			
1011		0101			
1100		0101			
1101		0101			
1110		0101			
1111		0101			

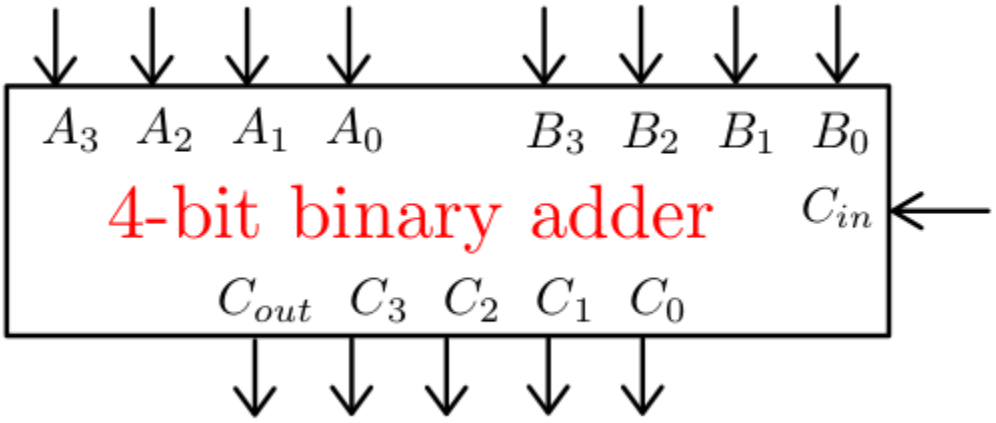
Now fill in the following table using the circuit from Task 1a. Each 4-bit number is a **signed** number. Circle those binary outputs where **signed** overflow has occurred.

Input A		**Input B**		**Output C = A + B**	
Binary	**Decimal**	**Binary**	**Decimal**	**Binary**	**Decimal**
0011		0000			
0011		0001			
0011		0010			
0011		0011			
0011		0100			
0011		0101			
0011		0110			
0011		0111			
1000		0101			
1001		0101			
1010		0101			
1011		0101			
1100		0101			
1101		0101			
1110		0101			
1111		0101			

Signed subtraction of 4-bit binary numbers (2's complement) using an integrated circuit

Task 2a [CLO2, 1 Mark]

The figure below shows a schematic for the 7483 IC, which adds two 4-bit numbers A = $A_3A_2A_1A_0$, B = $B_3B_2B_1B_0$ and a single bit C_{in} to produce a 5 -bit output $C_{out}C_3C_2C_1C_0$. By using only NOT gates show how this schematic can be **modified** to function like a 4-bit binary subtractor. (which subtracts two 4-bit **signed** integers X = $X_3X_2X_1X_0$ and Y = $Y_3Y_2Y_1Y_0$) to produce a 4-bit **signed** integer X – Y, by using the 2's complement method. (Draw your solution on **this** paper).



Task 2b [CLO1, 2 Marks]

The pin out of the 7404 and 7483 ICs is shown in the 'Pin out' section at the end. Implement your drawing in Task 2a using these ICs. Switches S0 to S3 provide the first 4-bit input X. Switches S4 to S7 provide the second 4-bit

output Y. LEDs L1 to L4 record the 4-bit binary output (X-Y) of the subtractor.

Task 2c [CLO 3, 1 Mark]

Now fill in the following table using the circuit from Task 2b. Each 4-bit number is a **signed** number. Circle those binary outputs where **signed** overflow has occurred.

Input X		**Input Y**		**Output X - Y**	
Binary	**Decimal**	**Binary**	**Decimal**	**Binary**	**Decimal**
0101		0000			
0101		0001			
0101		0010			
0101		0011			
0101		0100			
0101		0101			
0101		0110			
0101		0111			
1000		0011			
1001		0011			
1010		0011			
1011		0011			
1100		0011			
1101		0011			
1110		0011			
1111		0011			

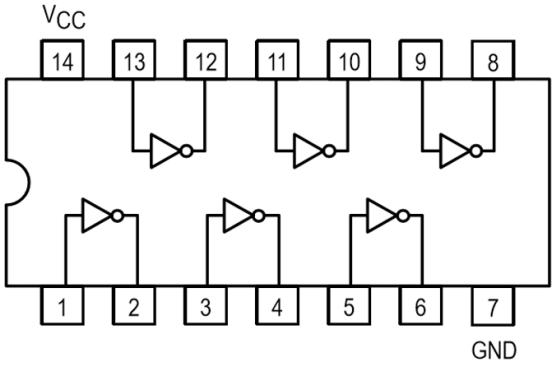
Interface a seven segment display unit with a 4-bit binary subtractor

Task 3 [CLO1, 2 Marks]

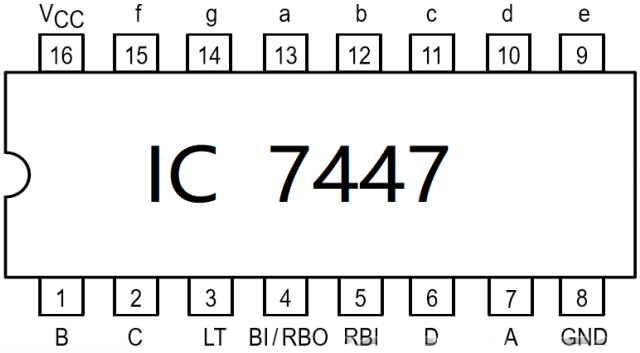
The pin out of a 7447 IC (seven segment display driver) is shown in the 'Pin out' section at the end. You have to interface the 4-bit output (X-Y) of the subtractor in Task 2b to a seven segment display using the 7447 IC. Remember to use the 220Ω resistors while connecting the pins of the seven segment display to its driver IC (7447), for limiting the current.

Pin out

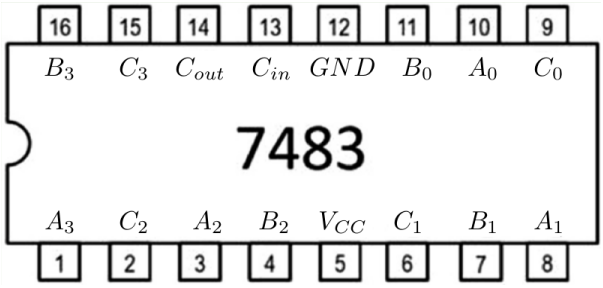
7404 IC



7447 IC



7483 IC



Seven-segment display

